

LUCAS SPAIN

Senior Software Engineer | Full-stack and Generative AI Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years building production full-stack applications and 3+ building agentic systems at enterprise scale, on a foundation of 10 years of Python and 6 years of React. I specialize in web apps and multi-agent orchestration (A2A, MCP), RAG LLM applications, and secure, low-latency microservices. I take AI and web products from prototype to customer-facing rollout, standardizing CI/CD, and teaching AI engineering to 100+ peers.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, Ruby

Gen AI & Agents: LLM Application Development, Agent-to-Agent (A2A) Protocol, Model Context Protocol (MCP), Multi-Agent Orchestration, Custom Actor Framework, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain/LangGraph, Context Management

Backend & APIs: FastAPI, Pydantic, REST APIs, GraphQL, Microservices, JWT/OAuth Authentication, WebRTC

Frontend: React 18, Micro-Frontends, Shared-State Architecture, Canvas/Media APIs

Cloud & DevOps: AWS, Terraform, CI/CD, GitHub Actions, Playwright, Docker, Kubernetes, ArgoCD

Observability & Data: OpenSearch, Grafana, Supabase, DataDog, PostHog, FullStory

Practices: Agile/Scrum, System Design, Test Automation, Secure SDLC, Technical Mentorship

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Internal AI — John Deere. *September 2025 – Present*

- Architected migration of the team's agent platform from LangGraph to the **Agent-to-Agent (A2A) protocol** on a custom actor framework, improving fault isolation, concurrency, and maintainability across multi-agent workflows.
- Delivered **Model Context Protocol (MCP) app integrations**, allowing for rich, interactive UI experiences in AI chat windows.
- Built a **unified dealer experience for AI**, consolidating fragmented assistant interfaces into a single, consistent entry point for AI capabilities across the dealer channel.
- Led **employee onboarding and instruction** for AI engineering, designing and teaching internal classes that upskilled new and existing engineers on agentic development practices.
- Engineered secure, low-latency FastAPI microservices for embeddable Gen-AI features.

Software Engineer — Channel AI Solutions — John Deere *December 2024 – September 2025*

- Co-led the **Virtual Sales Representative**: a React + HeyGen avatar that recommends equipment to customers; launched live in a dealership with a rollout plan targeting 50+ locations.
- Built a LangGraph-powered **Dealer Assistant** integrating the Customer Data Product; reduced warranty-lookup steps from 7 to 1.

Software Engineer — Order Management / Build & Price (ITDP Rotation) — John Deere. *Jun 2022 – Dec 2024*

- Unified three React micro-frontends (~250K LOC) into a shared-state Build & Price application, improving first-paint performance by 37%.
- Launched the organization's first autonomous CI/CD pipeline (GitHub Actions + AWS) with a multi-environment lint → unit → Playwright → deploy flow reused by 5+ teams.
- Developed the “Francois” bot for resilient workflow retries and auto-rollback, reducing CI spot instance failures by 100%.
- Discovered and patched a critical CSRF vulnerability in Sales Center; standardized the mitigation across six teams.

SELECTED PROJECTS

- **Gatsby Home Assistant** — LangGraph voice agent (Parakeet + Piper) controlling smart-home devices.
- **faster-whisper-recognition** — Open-source Wyoming satellite for fast transcription and speaker identification.
- prideinmotion.com — React + Supabase platform for the Quad Cities' largest LGBTQ-inclusive recreational sports league, serving 200+ weekly visitors. Sole owner, builder, and maintainer.

LEADERSHIP & OUTREACH

- Instructor, John Deere AI Hackathon prep — deliver regular sessions ranging from 1-4 hours on building AI agents to ~100 engineers on average.
- Mentor to a University of Iowa Computer Science student, providing ongoing monthly technical guidance.
- Leadership role with Pride in Motion (Quad Cities recreational sports nonprofit 501(c)(3)); own and maintain its web platform as a volunteer.

EDUCATION

M.S., Computer Science — University of Iowa · GPA 3.85 - 2022